

Reducing the player's base max health using `/attribute` does not always reduce their current health

MC-16345 Data Packs

Status	Resolved / Fixed
Affected	Snapshot 13w21a, Minecraft 1.6.1, Minecraft 1.6.2, Minecraft 1.7.4, Minecraft 14w02c, Minecraft 14w03a (+87 more)
Fixed in	24w19a
Reported	2013-05-23

STEPS TO REPRODUCE

1. Create or load a world and enter Survival mode.
2. Run `/gamerule naturalRegeneration false``.
3. Run `/scoreboard objectives add health health``.
4. Run `/scoreboard objectives setdisplay sidebar health``.
5. Run `/attribute @s minecraft:generic.max_health base set 30``.
6. Run `/effect give @s minecraft:instant_health 1 20 true`` to restore current health to full.
7. Run `/attribute @s minecraft:generic.max_health base set 20``, then run `/attribute @s minecraft:generic.max_health base set 1``.
8. Compare the health value shown in the sidebar with the hearts displayed on screen.
9. If the sidebar value does not exceed the visible hearts, run `/attribute @s minecraft:generic.max_health base set 30`` and `/effect give @s minecraft:instant_health 1 20 true``, then repeat steps 7-8.
10. Observe that the sidebar reports more current health than the visible hearts, showing that reducing the base maximum health did not always reduce the current health accordingly.

OBSERVED BEHAVIOUR

Actual behavior

Changing the player's `minecraft:generic.max_health`` base value with `/attribute`` does not consistently clamp or otherwise update the player's current health.

For example, when the player has 30 health and their base max health is changed from 30 to 20 or 1, the server can continue to store the player's current health as 30. The `health`` scoreboard objective then reports a value greater than the value returned by:

```
``mcfunction
attribute @s minecraft:generic.max_health get
``
```

This leaves the player with a current health value above their calculated maximum health. The issue is intermittent: in some attempts, reducing the base max health updates the current health as expected, while in others it does not.

The client is still informed of the changed maximum health, so its heart display can become inconsistent with the server-side health value. In particular, after the maximum health is reduced, the scoreboard can show more health than is represented by the player's hearts. Restoring the base max health can update the maximum-health display without restoring the missing red hearts.

Taking damage or receiving healing causes the current health to be synchronized again, after which the discrepancy disappears.

EXPECTED BEHAVIOUR

When the player's base maximum health is reduced with `/attribute``, their current health should be reduced as necessary so it never exceeds the new maximum. The server and client should remain synchronized, with the scoreboard and displayed hearts showing the same valid health value. Increasing the maximum health again should not leave stale or inconsistent current-health values.

ENVIRONMENT

Minecraft Java Edition Versions: 13w21a, 1.6.1, 1.6.2, 1.7.4, 14w02c, 14w03a, 14w03b, 1.7.10, 14w30c, 14w31a, 14w33c, 14w34a, 14w34b, 14w34c, 14w34d, 1.8-pre1, 1.8.2-pre4, 1.8.8, 15w37a, 1.10.2, 16w41a, 16w43a, 16w44a, 1.11 Pre-Release 1, 1.11, 16w50a, 1.11.1, 1.11.2, 17w06a, 17w13b, 17w15a, 17w16b, 17w17a, 17w18b, 1.12 Pre-Release 2, 1.12.2, 18w15a, 1.13.1, 1.14.2, 1.14.4, 19w45b, 1.15.2, 20w17a, 20w18a, 20w19a, 20w20b, 20w21a, 1.16 Pre-release 1, 1.16 Pre-release 2, 1.16 Release Candidate 1, 1.16, 1.16.1, 1.16.2 Pre-release 1, 1.16.2 Pre-release 2, 1.16.2 Release Candidate 1, 1.16.2, 1.16.4, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w07a, 21w11a, 21w14a, 21w15a, 21w16a, 21w17a, 1.17 Pre-release 1, 1.17, 1.17.1, 21w39a, 21w43a, 1.18, 1.18.1, 1.18.2, 22w18a, 1.19, 1.19.2, 1.19.3 Release Candidate 3, 1.19.3, 23w03a, 1.19.4, 23w18a, 1.20.1, 23w31a, 23w32a, 23w35a, 1.20.2 Pre-release 1, 23w44a, 24w04a, 1.20.5, 1.20.6 (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When the player's base max health is reduced using `/attribute`, their current health (stored on the server) is not always reduced accordingly. (There have been reports in the past that this bug occurs whenever the player's total max health is reduced, including using attribute modifiers, but as of the 1.20.2 snapshots I can only reproduce this by modifying the base value using `/attribute`.)

Reproduction

Setup

This bug occurs inconsistently and can be hard to detect when it occurs unless you're paying close attention, so it's preferable to have a dedicated testing setup. Let's start with a few of commands:

```
/gamerule doImmediateRespawn true
```

```
/gamerule keepInventory true
```

```
/gamerule naturalRegeneration false
```

```
/spawnpoint @s ~ ~ ~
```

Now let's make our current health value visible in the sidebar:

```
/scoreboard objectives add health health
```

```
/scoreboard objectives setdisplay sidebar health
```

```
/kill @s
```

And, for extra visibility, let's make particles emanate from ourselves when our current health exceeds our max health. First we run the following:

```
/scoreboard objectives add temp dummy
```

Then we put the following in a powered repeating command block:

```
execute as @p store result score @s temp run attribute @s minecraft:generic.max_health get 1.0
```

And the following in an unconditional, always-active chain command block pointing away from the repeating one:

```
execute as @p at @s if score @s health > @s temp run particle minecraft:glow ~ ~ ~ 0.2 0 0.2 0.2 10
```

Now let's set up some impulse command blocks which can cause the bug to occur. Place a button on each one, and make sure they aren't adjacent to each other.

```
attribute @p minecraft:generic.max_health base set 1
```

```
attribute @p minecraft:generic.max_health base set 20
```

```
attribute @p minecraft:generic.max_health base set 30
```

Finally, we need a command block to heal ourselves to full in case we've failed to repro and want to try again. Place a button on this one too.

```
effect give @p minecraft:instant_health 1 20 true
```

Reproducing the bug

Go into survival mode. Using the command blocks you just set up, set your base max health to 30 and heal yourself to full. Try reducing your max health, first to 20, then to 1. I

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-16345>